Daniel Medynski, Ph.D.

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Research | Stem Expert | Biochemistry | Recombinant Protein Purification | Scientific Writing | Prompt Engineering to guide AI models | In vitro Protein Refolding | Protein Chemistry | Molecular Biology | Protein Pegylation | Bio-Assay Development | Data Analysis

PROFESSIONAL SUMMARY

Ph.D. scientist and USPTO-registered Patent Agent with 30+ years of experience spanning protein chemistry, molecular biology research, biotechnology patent analytics, and AI/LLM training. Combines deep technical expertise in protein science, genomics, and pharmacology with a proven track record in NIH grant writing, peer-reviewed publications, and scientific communication. Currently building applied AI skills in prompt engineering and LLM evaluation. Seeking fully remote roles in AI training, scientific/technical writing, grant writing, or patent analytics within the biotechnology sector. Immediate Availability November 2026 up to 30 hours/week.

CORE COMPETENCIES

- AI/LLM Training & Evaluation — multimodal prompt engineering, scientific accuracy challenge, rubric writing
- Biotechnology Patent Analytics — prior art, invalidity/infringement/clearance/FTO/landscape searches; PCT/ISA evaluation
- Grant Writing & Management — Principal Investigator on 4 NIH SBIR grants (~$950K); budget management
- Scientific & Technical Writing — peer-reviewed publications, patent applications, regulatory opinions
- Protein Science — expression, refolding, purification (HPLC/FPLC), immunoassay development
- Domain Expertise — Biochemistry, Molecular Biology, Microbiology, Immunology, Genomics, Proteomics, Pharmacology, Stem Cell Biology, Nanotechnology, Bioinformatics, NGS, Gene Editing

PROFESSIONAL EXPERIENCE

AI Tutor and Trainer (Subject Matter Expert, Biology)

Micro1 (www.micro1.ai)| Remote Jul 2026-Oct 2026

- Served as Ph.D.-level SME to train, challenge, and evaluate LLM responses in biology and biotechnology domains.
- Challenge reasoning ability for 8 different LLM responses for each task and evaluate accuracy of scientific responses. Write prompts that defeat LLMs.

AI Fellow & Trainer (Subject-Matter Expert, Biology)

Handshake AI — MOVE Program | Remote *Jul 2025 – Nov 2025*

- Served as Ph.D.-level SME to train, challenge, and evaluate LLM responses in biology and biotechnology domains.
- Developed multimodal prompts (text + image) to rigorously test LLM scientific reasoning and factual accuracy, focused on the domain of biology.
- Authored evaluation rubrics for systematic assessment of model outputs.

Senior Patent Analyst / Patent Analyst (Patent Scientist)

Cardinal Intellectual Property | Remote *Jul 2007 – Nov 2023*

- Conducted prior art, invalidity, infringement, clearance, and FTO searches across international biotechnology patent portfolios for 16+ years.
- Evaluated patent applications as an International Search Authority (ISA) examiner under the PCT, under contract with the USPTO.

- Produced detailed technical written opinions on patentability merit and claim scope across biochemistry, molecular biology, proteomics, bioinformatics, stem cell biology, immunology, pharmacology, next generation sequencing, and genomic editing.

Senior Scientist & Group Leader
Proteomtech Inc. | Emeryville, CA *Aug 2002 – Aug 2006*

- Led protein science R&D with focus on protein chemistry (pegylation), refolding, and purification. Enzymatic analysis using spectroscopic assays.
- Awarded Principal Investigator status on 4 NIH SBIR grants totaling ~$950,197; co-managed project budgets.
- Supervised team members, organized research meetings, and authored peer-reviewed publications and patent applications.

Scientist II
Diadexus, Inc. | South San Francisco, CA *Nov 1999 – Jul 2002*

- Conducted research in protein science, proteomics, and immunoassay development; supervised 2 FTEs.

Scientist
Roche Bioscience — Inflammatory Disease Unit | Palo Alto, CA *Feb 1996 – Oct 1999*

- Designed, optimized, and scaled protein purification procedures using HPLC/FPLC; supervised 3 FTEs.

Scientist
Khepri Pharmaceuticals | South San Francisco, CA *Feb 1995 – Dec 1995*

- Research in enzymology of cysteine proteases (e.g. cathepsin K) including protein expression and purification.

Scientist — Analytical Development
Amcell Corporation | Sunnyvale, CA *Feb 1993 – Feb 1995*

- Research in immunochemistry applications related to stem cell isolation using the CD34 biomarker.

Visiting Scientist
Dana Farber Cancer Institute & Harvard Medical School — Dept. of Pathology | Boston, MA *1990*

Freelance Science Writer & Columnist
Biotechnology & Medical Publications *1991 – 1993*

- Wrote columns and articles on scientific, medical, and biotechnology topics for industry publications.

EDUCATION

Postdoctoral Studies (1983-1989) — Molecular Biology, Microbiology & Protein Science
University of California, San Francisco (UCSF)

Ph.D., Molecular Pharmacology (1977-1983)
Harvard Medical School, Boston, MA

B.A., Biology (Molecular & Cellular), Minor in Chemistry (1971-1975)
Haverford College, Haverford, PA

CERTIFICATES & LICENSES

- USPTO Patent Agent License — Reg. No. 62,671 (April 2008, non-expiring)
- Google AI Professional Certificate — Google/Coursera/(August 2026)
- Google Prompting Essentials — Google / Coursera (February 2026)
- Python Data Structures — University of Michigan / Coursera (February 2026)
- Programming for Everybody (Python) — University of Michigan / Coursera (December 2025)
- AI for Everyone Specialization — IBM / Coursera (May 2025)

PATENTS

- US2007/0218535 A1 — Methods for Production of Recombinant alpha1-Antitrypsin (Lin & Medynski)
- WO/2007/062270 A2 — Methods for Production of Recombinant alpha1-Antitrypsin (Lin & Medynski)
- GB 2446550 A — Methods for Production of Recombinant alpha1-Antitrypsin (Lin & Medynski)
- WO/2008/054592 A2 — Methods for Production of Recombinant Plasminogen and Plasmin Polypeptides (Lin & Medynski)
- US2010/0144622 A1 — Methods for Production of Recombinant Plasminogen and Plasmin Polypeptides (Lin & Medynski)

SELECTED PEER-REVIEWED PUBLICATIONS

1. Zhu W, Deng M…..Medynski, D….Oxidation-resistant and thermostable forms of alpha-1 antitrypsin from *Escherichia coli* inclusion bodies. FEBS Open Bio Vol 8 No 10 Pages 1711-1721 (2018). Link: https://febs.onlinelibrary.wiley.com/doi/10.1002/2211-5463.12515

2. **Medynski D**, Tuan M, Liu W , Wu S, X. Lin. Refolding, purification, and subsequent functional enzymatic activation of miniplasminogen and microplasminogen isolated from *E. coli* inclusion bodies. Protein Expression and Purification 52: 395-402 (2007). Link: https://doi.org/10.1016/j.pep.2006.10.012

3. Hou W, **Medynski D**, Wu S, Lin X, Li LY. VEGI-192: a new isoform of TNFSF15, specifically eliminates tumor vascular endothelial cells and suppresses tumor growth. Clinical Cancer Research 11(15):5595-5602 (2007). Link: https://doi.org/10.1158/1078-0432.CCR-05-0384

4. Shujath Ali, Yongming Sun, Carrie Drumright, Luis Carretero, Christina Tran, Anton Nguyen, **Dan Medynski**, Yan Liu, Vu Viet Hoang, Brandon Liang, Herve Recipon, Robert Cafferkey, Roberto Macina. Genomic Approaches to the Development of Prostate Cancer Diagnostics. Nature Genetics 27, 38 (2001). (abstract)

5. **D. Medynski**. /Biotools: Synthetic Peptide Combinatorial Libraries: Using Applied Molecular Evolution to Accelerate Drug Discovery. Bio/Technology 12: 709-710(1994). Link: https://doi.org/10.1038/nbt0794-709

6. **D. Medynski**. /Biotools: Phage Display: All Dressed-Up and Ready to Role. Bio/Technology 12: 1134-1136(1994). Link: https://doi.org/10.1038/nbt1194-1134

7. **D. Medynski**. Genetic Approaches to Protein Structure and Function: Point Mutations as Modifiers of Protein Function by D. Medynski. Bio/Technology 10: 1002-1006(1992). Link: https://doi.org/10.1038/nbt0992-1002

8. **D. Medynski** and K.N. Tan. Genetic Approaches to Protein Structure and Function: II. Rapid Generation and Identification of Multiple Mutants at Defined Protein Sites Utilizing an E. coli Colony Color Screen and tRNA Amber Suppressors. 1991.

9. C. Van Dop, **D. Medynski**, and L. Apone. Nucleotide Sequence for the Alpha Subunit of Transducin cDNA isolated from the Human Eye. Nucl. Acids Res.17: 4887(1989).

10. HR Bourne, GF Casperson, N Walker, K Sullivan & **DC Medynski**: Adenylate cyclase and GTP-binding proteins in yeast and vertebrate cells. IN: Yeast Cell Biology (Proceedings of 1985 UCLA Symposium), Vol. 33, J. Hicks (ed.), Alan R. Liss Publishers, New York, pp.81-100, 1986.

11. **D. Medynski**, K. Sullivan, F. Chang, D. Smith, B. Fung, P.H. Seeburg, and H. Bourne. Amino Acid Sequence of the Alpha Subunit of Transducin from the cDNA Sequence. Proc. Nat. Acad. Sci. 82: 4311-4315(1985). Link: https://doi.org/10.1073/pnas.82.13.4311

12. Van Dop C, **Medynski D**, Sullivan K, Wu A, Fung B, Bourne H. Partial cDNA Sequence of the Gamma Subunit of Transducin. Biochem. Biophys. Res. Commun. 124:250–255 (1984).

13. Bourne H, **Medynski D**, Van Dop C, Sullivan K, Chang F. Genetic and Functional Studies of Pertussis Toxin Substrates. IN: Pertussis Toxin, pp. 167–184 (1985).

14. Cohen J, Correll L, Dreyer E, Kuisk I, **Medynski D**, Strnad N. Interaction of Local Anesthetics with Torpedo Nicotinic Acetylcholine Receptors. IN: Molecular and Cellular Mechanisms of Anesthesia, pp. 314–340 (1985).

15. Cohen J, **Medynski D**, Strnad N. Interaction of Local Anesthetics with Nicotinic Acetylcholine Receptors. IN: Effects of Anesthesia, pp. 53–64, American Physiological Society (1985).

16. **Medynski D**. Histrionicotoxin and Aromatic Amine Interactions with Nicotinic Acetylcholine Receptors from Torpedo. Ph.D. Thesis, Harvard University (1983). Dissertations Abstracts 44:1800B.

17. Cohen J, Boyd N, Jeng A, **Medynski D**, Neubig R, St. John P. Permeability Control by Torpedo Acetylcholine Receptors. IN: Psychopharmacology and Biochemistry of Neurotransmitter Receptors, pp. 47–62 (1980).